Dollar-neutral vs. beta-neutral

Why our long-short portfolio uses equal-dollar weighting and what that actually buys (and costs) us.

Project ml-factor-investing-2026 | Person B (Alpha Model) | 2026-05-22

The question, in one sentence

Person B's strategy mechanics report describes the portfolio as *long $1 in the top 20% of stocks, short $1 in the bottom 20%, equal weight within each leg*. That makes it **dollar-neutral**. A correctly informed reader pointed out that **dollar-neutral is not the same as market-neutral**: if the long-leg stocks have higher market beta than the short-leg stocks (which they usually do in a winners-vs-losers basket), then equal dollars leaves you net-long the market. The technically tidier approach is **beta-neutral** weighting, where you size the two legs so that the beta-weighted exposures cancel. So why are we not doing that?

The one-paragraph answer

Four reasons, in priority order: (1) it is what the framework and the Gu-Kelly-Xiu (2020) paper this project replicates specify; (2) it isolates the cross-sectional stock-picking skill we are actually trying to measure, instead of mixing it with the success or failure of a beta hedge; (3) beta is itself a noisy quantity that you would be hedging against a moving target; (4) some of what looks like alpha in factor strategies *is* beta-loaded by construction, and hedging it away removes part of the very signal you are trying to capture. The honest cost is that our portfolio is not exactly market-neutral — net beta is probably around +0.2 to +0.4 — which the final report will disclose, and which we propose to quantify with a sensitivity check.

1. Definitions, precisely

Three increasingly demanding ways to construct a long-short basket. Each one removes a different source of risk.

1.1 Dollar-neutral (what we do)

Long L dollars, short S dollars, with L = S. The long and short legs have the same total dollar size. Net cash position after taking the trades is unchanged from before.

Math:

```
L = S
Portfolio return = (long-leg return × L − short-leg return × S) / NAV
```

This makes **no claim** about market exposure. If both legs happened to be 100% utility stocks, both would move with the utility sector. If the long leg is tech and the short leg is utilities, the portfolio is implicitly long-tech / short-utilities.

1.2 Beta-neutral (the alternative)

Size the legs so that the **beta-weighted** exposures match. If the long leg has higher average beta than the short leg, you either short more dollars (relative to longs) or long fewer dollars (relative to shorts) so that the beta-weighted dollar amounts are equal in magnitude.

Math:

```
L · β_L = S · β_S
where β_L = weighted-avg market beta of the long leg
β_S = weighted-avg market beta of the short leg

Solving: S / L = β_L / β_S
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A portfolio constructed this way has a net beta of zero *on average*, against the chosen market benchmark, over the estimation window the betas were computed on.

1.3 Mean-variance optimal (the textbook ideal — not feasible here)

Solve for weights that maximise expected return per unit of portfolio variance, given the model's predictions as the return estimates and an estimated covariance matrix. Beautiful in theory, brittle in practice — the covariance matrix has ~$500^2 \approx 250{,}000$ entries you have to estimate, and the optimal weights amplify any error. Not what GKX uses, not what we'll do here.

2. A concrete dollar-by-dollar example

To make the difference tangible, here's a hypothetical month with realistic numbers.

2.1 The setup

Suppose at month-end the model has selected:

- **Long leg:** 100 stocks, mostly momentum winners (growthy, smaller-cap-ish, recent strong returns). Beta-weighted average market beta = **β_L = 1.20**.

- **Short leg:** 100 stocks, mostly recent losers and defensive low-beta names. Beta-weighted average market beta = β_**S = 0.90**.
- **NAV:** $1,000.

(These betas are realistic — in factor strategies the top-decile vs. bottom-decile beta spread typically ranges from 0.2 to 0.4.)

2.2 Dollar-neutral weighting (what we do)

Equal $1000 long, $1000 short. Gross exposure $2000 (2x leverage). Net dollar exposure $0.

If the broad market rises 10% over the month (and stocks track their betas perfectly), the leg P&L is:

```
Long-leg return = +β_L × 10% = +12.0% → +$120
Short-leg P&L; = −β_S × 10% = −9.0% → −$90 (we're SHORT)
Wait — that's wrong. If we're SHORT the short leg, we LOSE money
when the short leg goes up. Let me redo it more carefully:

Long-leg position rises by β_L × 10% × $1000 = +$120
Short-leg position rises by β_S × 10% × $1000 = +$90
But we're SHORT the short leg, so its rise is our LOSS = −$90

Total P&L; from beta alone = +120 − 90 = +$30
On a $1,000 NAV, that's +3% pure beta drift.
```

What this means: If the model is no better than random and the market rises 10%, we'd still show +3% return. That's not alpha — it's a hidden +0.30 net beta times a +10% market move. In a bull market we'd over-credit the model. In a 10% sell-off we'd see –3% and wrongly blame the model.

2.3 Beta-neutral weighting (the alternative)

Same gross exposure of $2,000, but redistribute the dollars so the beta-weighted legs match.

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Constraint: L + S = $2,000 (same gross exposure)
Constraint: L · β_L = S · β_S (beta-neutral)

Substituting β_L=1.20, β_S=0.90:
L · 1.20 = (2000 − L) · 0.90
1.20·L = 1800 − 0.90·L
2.10·L = 1800
L = $857 → long position scaled DOWN
S = $1,143 → short position scaled UP

Verify: 857 · 1.20 = 1,028.6
1,143 · 0.90 = 1,028.6 ✓ beta-weighted legs match
```

Now the same +10% market move produces:

```
Long-leg P&L;: +β_L × 10% × $857 = +$103
Short-leg P&L;: −β_S × 10% × $1143 = −$103
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```
Net P&L; from beta alone = $0 ✓
```

What this means: The portfolio's P&L is now (essentially) independent of the market direction. Whatever return we observe comes from the cross-sectional spread between the legs, not from the market rising or falling.

So why doesn't every long-short fund just do this?
Because the β values you used to size the legs are themselves estimates. If your β_L was actually 1.30 (you measured 1.20 but were wrong), then the portfolio retains a +0.10 net beta even after the "hedge" — you just lost confidence about how much. See section 4.

3. The four reasons we chose dollar-neutral anyway

3.1 The framework and GKX (2020) explicitly use it

Project Framework section 6.1 specifies "long/short the top/bottom k stocks within each sector, equal-weighted within each leg." The Gu-Kelly-Xiu (2020) paper, whose methodology we are replicating, uses the same construction throughout its empirical section. Replicating their methodology is part of what the project is supposed to do. Deviating to beta-weighting would require an explicit justification in the report — which we can do, but then we'd have two methodologies running side by side instead of one canonical one.

3.2 Isolating skill from a beta hedge

There is a clean separation of concerns in our project:

Question we're trying to answer	Tool
Did the ML model pick winners over losers?	Cross-sectional ranking metrics (IC, R^2) and the spread between top-decile
How much risk did the strategy take?	Sharpe ratio, max drawdown — measured on the dollar-neutral portfolio
Should the strategy take more or less risk in different market environments?	Person C's regime overlay — leverage, breadth, and threshold dialled by t

Beta-neutralisation would add a fourth concern: **did the beta hedge succeed?** If the strategy looks good, was it the model picking winners, or was it the beta hedge effectively timing the market by accident? Disentangling these post-hoc is annoying. Keeping the portfolio dollar-neutral means the test of "did the model work?" is clean: any return either came from the spread or from beta drift, and we can measure the beta drift component separately to subtract it.

3.3 Beta estimation is itself noisy

Beta is not a constant — it has to be estimated from past data. Common approaches and their per-stock noise:

Estimation method	Per-stock standard error of β
Rolling 12 months of monthly returns	≈ 0.30 - 0.40
Rolling 36 months of monthly returns	≈ 0.15 - 0.20
Rolling 60 months of monthly returns	≈ 0.10 - 0.15
Daily-data 1-year regression	≈ 0.10 - 0.15

Take the 36-month case as representative. A stock's β estimate has a 1-sigma error of ~0.18. For a single name that's a ~15% relative error around β=1.20. When you average across 100 names in a leg, the leg-level β_L estimate gets tighter (~0.018 if the noise were independent, but it isn't because of common factor exposures — call it 0.05 in practice).

So β_L=1.20 might actually be 1.15 or 1.25, and we wouldn't know. Hedging against a noisy estimate gets you most of the way to beta-neutral, but not exactly there — and crucially it introduces estimation noise into the portfolio that wasn't there before.

Worse: beta is regime-dependent. A bank's beta in calm markets is ~0.9; in a financial crisis it can spike to 1.8. Rolling-window beta lags this by half the window length. The very moment you need the hedge most (a crash) is when your beta estimate is most wrong.

3.4 Some of the "alpha" IS beta-loaded by construction

This one is subtle but important. Momentum, the strongest single feature in our model, is not a pure alpha: stocks that have gone up tend to keep going up partly **because** they have higher beta in trending markets. The high-beta loading is *itself* part of the momentum effect.

If we beta-hedge our long-momentum / short-momentum portfolio, we remove some of the very return we are trying to capture. Empirical estimates suggest 20-40% of the conventional momentum premium is actually a beta tilt; hedging removes that component but keeps the residual stock-specific momentum (which is real but smaller). For a course project measuring the model's ability to *pick* stocks, you may not want to obscure this — let the strategy capture the full effect, report the net beta separately so the reader knows.

> Empirical aside: in our actual numbers, the tuned XGBoost on the 2019-2024 test window produces a Sharpe of +0.53 dollar-neutral. A beta-neutral version would likely deliver around +0.40 - +0.50 — slightly worse (because we'd be hedging out some real return) but with a genuinely zero net market exposure. We'll quantify this if we run the sensitivity check.

4. What we give up by choosing dollar-neutral

Three real costs, none catastrophic, all disclosable in the report.

4.1 The portfolio is not exactly market-neutral

Our PORTFOLIO is dollar-neutral, but its NET market beta is non-zero. In factor strategies the long leg systematically loads on higher-beta names (momentum, growth, smaller caps); the short leg loads on lower-beta names (defensives, large value, utilities). Empirical net beta for top-vs-bottom decile strategies usually sits in [+0.2, +0.5].

What this implies:

- In a strong bull market, our Sharpe overstates pure stock-picking skill (we're getting paid for the beta drift too).
- In a strong bear market, our drawdowns are deeper than a true market-neutral portfolio would show.
- Our 2019-2024 test window included one large bull (2020-2021, 2023-2024) and one significant drawdown (2022). On balance the beta drift probably helped the Sharpe slightly.

4.2 "Market-neutral" is technically the wrong word

The strategy mechanics report previously used "market-neutral by construction". That phrasing is sloppy — it should say "dollar-neutral by construction; net market beta is small but not exactly zero." That language will be corrected in the next regeneration of that PDF.

4.3 A regulator or risk-manager would want both numbers

If this were a real trading desk, the risk team would compute and report BOTH the dollar-neutral and beta-neutral versions. The first is the constructed portfolio; the second is the "market-risk-equivalent" version a risk officer cares about. For a course project the dollar-neutral number suffices, but the report should note that the beta-neutral comparison is the standard production-grade additional check.

5. What we propose to add (the sensitivity check)

Two-hour task. Strengthens the report without changing the headline number. Implementation lives in a new notebook (`notebooks/personb/05_beta_neutral_check.py`) that operates entirely on the predictions already produced by Phase 3c — no changes to Bowen's backtest engine.

5.1 The steps

#	Step	Detail
1	Estimate per-stock β	Rolling 36-month OLS of stock return on S&P 500 return, lagged 1 month so we use only
2	Compute leg betas	At each rebalance, β_L = average β of the top-decile stocks; β_S = average β of the botto
3	Re-weight legs	Replace equal-$1000/equal-$1000 with $L*\beta_L = S*\beta_S$ under the same total gross expos
4	Re-realise returns	Compute the portfolio return series under the new weights, month by month, on the same
5	Compare metrics	Side-by-side: Sharpe, ann return, max drawdown, net beta (should be ≈0 for the beta-neu

5.2 What the report will show

A two-row table:

Construction	Net β	Sharpe	Ann return	Max DD	Comment
Dollar-neutral (canonical)	+0.3	+0.53	+4.86%	–14.0%	GKX-style, framework-prescribed
Beta-neutral (sensitivity)	≈ 0	+0.4X	+3.X%	–1X%	Strips out beta drift; "pure" cross-sectional

(The numbers in row 2 are placeholders; the real values come out of the sensitivity check itself.)

5.3 The narrative for the report

"We construct the portfolio dollar-neutral, following the GKX (2020) replication convention. The strategy's headline Sharpe of +0.53 includes a small net long-market exposure ($\beta \approx +0.3$) from the long-decile loading on higher-beta names. We re-run the same predictions under a beta-neutral weighting (re-scaling legs so $\beta_L \cdot L = \beta_S \cdot S$) and find Sharpe of +0.4X with net beta ≈ 0, confirming the residual cross-sectional skill is approximately X% of the dollar-neutral headline."

> **The neat thing about this framing:** it doesn't undermine our main result — it strengthens it. We're not saying "the headline Sharpe is fake." We're saying "here's how much of it survives the most demanding hedge a risk-officer would ask for, and the answer is most of it."

6. When the decision would flip

Conditions under which we'd switch beta-neutral to the *primary* construction (not just a sensitivity check):

- If the sensitivity check reveals our net beta is large (say, +0.6 or higher), so the headline Sharpe is materially exaggerated by beta drift. Threshold for concern: net beta > +0.5 over the test window.

- If we were running this strategy at a fund where the investment mandate is "market-neutral". For a course project this doesn't apply.
- If GKX (2020) were superseded by a more recent paper that shifted methodology to beta-neutral. Not the case as of writing — most academic factor work still uses equal-weighted or value-weighted decile baskets.
- If we had a much shorter test window where beta drift would dominate any cross-sectional signal. Our 5-year test window is long enough to absorb beta drift in the noise.

7. Phrasing in the final report (short paragraph)

Suggested wording for the methodology section, ready to paste:

> *The portfolio is constructed as a long-short basket: long the top-20% stocks by predicted return, short the bottom-20%, equal-weighted within each leg, dollar-neutral by construction (matching the convention of Gu, Kelly & Xiu, 2020). This does not guarantee market-neutrality: the long leg systematically loads on higher-beta names than the short leg, so the realised portfolio has a small positive net market beta (approximately +0.3 in our test window). We report a beta-neutral sensitivity check in Appendix X showing that the Sharpe ratio remains positive after this exposure is hedged out, confirming that the strategy's risk-adjusted return is predominantly cross-sectional skill rather than disguised market timing.*

This is roughly 130 words. Sits cleanly in the methodology section's portfolio-construction paragraph.

Appendix A. We actually measured the net beta

After this report was drafted, Phase 5b ran an OLS regression of the canonical Phase-3c portfolio returns on the S&P; 500's monthly return over the full 2019-2024 test window. 72 months of data, Newey-West HAC standard errors with 6 lags.

A.1 Result

Model	β	HAC SE	t-stat	p-value	Ann. α	R^2 to mark
Lasso	-0.005	0.054	-0.09	0.93	+0.22%	0.000
XGBoost (canonical)	+0.046	0.040	+1.15	0.25	+4.69%	0.008
NN	+0.134	0.078	+1.71	0.09	+0.52%	0.040

The canonical XGBoost portfolio's net beta is +0.05, NOT +0.2 to +0.4 as the literature would suggest. The t-stat of 1.15 means we cannot reject $\beta = 0$ at any conventional significance level. Market explains only 0.8% of the portfolio's return variance. The +5.16% annualised return is almost entirely captured by the +4.69% annualised alpha — i.e., it is genuine cross-sectional skill, not disguised market timing.

A.2 Why the worry was unfounded — Layer 1 does it implicitly

The four reasons we gave for choosing dollar-neutral are still valid. What we under-appreciated: the Framework's Layer-1 step (sector-relative percentile ranks) silently does most of the beta-hedging work even though we never asked it to.

Walk through the chain:

- Every feature, including momentum, becomes a within-sector rank in [0, 1] before XGBoost sees it.
- When XGBoost predicts, its top-decile picks are *winners relative to their sector*, not absolute highest-momentum names globally.
- Across the 11 GICS sectors, the long basket spreads ~9-10 names per sector, and so does the short. The basket is sector-balanced by construction.
- Within a sector, the beta gap between the top-decile and bottom-decile names is small (~0.1-0.2). When the long-vs-short beta gap is averaged across 11 such sector pairs with near-equal weights, the gross book-level gap collapses to essentially zero.

So the dollar-neutral construction **plus the Layer-1 sector-rank step** is empirically beta-neutral. The naive worry ($\beta \approx +0.3$) applies to a non-sector-balanced top-vs-bottom-decile strategy on raw features — which is not the construction we actually use.

A.3 Implication for the methodology section of the report

Replace the previously-suggested paragraph with: "The portfolio is constructed as a long-short basket: long the top-20% stocks by predicted return, short the bottom-20%, equal-weighted within each leg, dollar-neutral by construction (matching the convention of Gu, Kelly & Xiu, 2020). We regress the realised portfolio returns on the S&P 500 monthly return over the test window and find $\beta = +0.046$ ($t = 1.15$, $p = 0.25$, $R^2 = 0.008$), indicating the portfolio is empirically market-neutral despite not being explicitly beta-hedged. The sector-relative feature ranking (Layer 1) and the resulting sector-balanced long-short basket appear to neutralise the market

> *exposure implicitly."*

That paragraph is the strongest possible defence: we did not skip a hedge — we measured the thing the hedge would have removed, and found it was already zero.